

# Homework 2: Time Series

Mateus Auza Cruz (28732200)

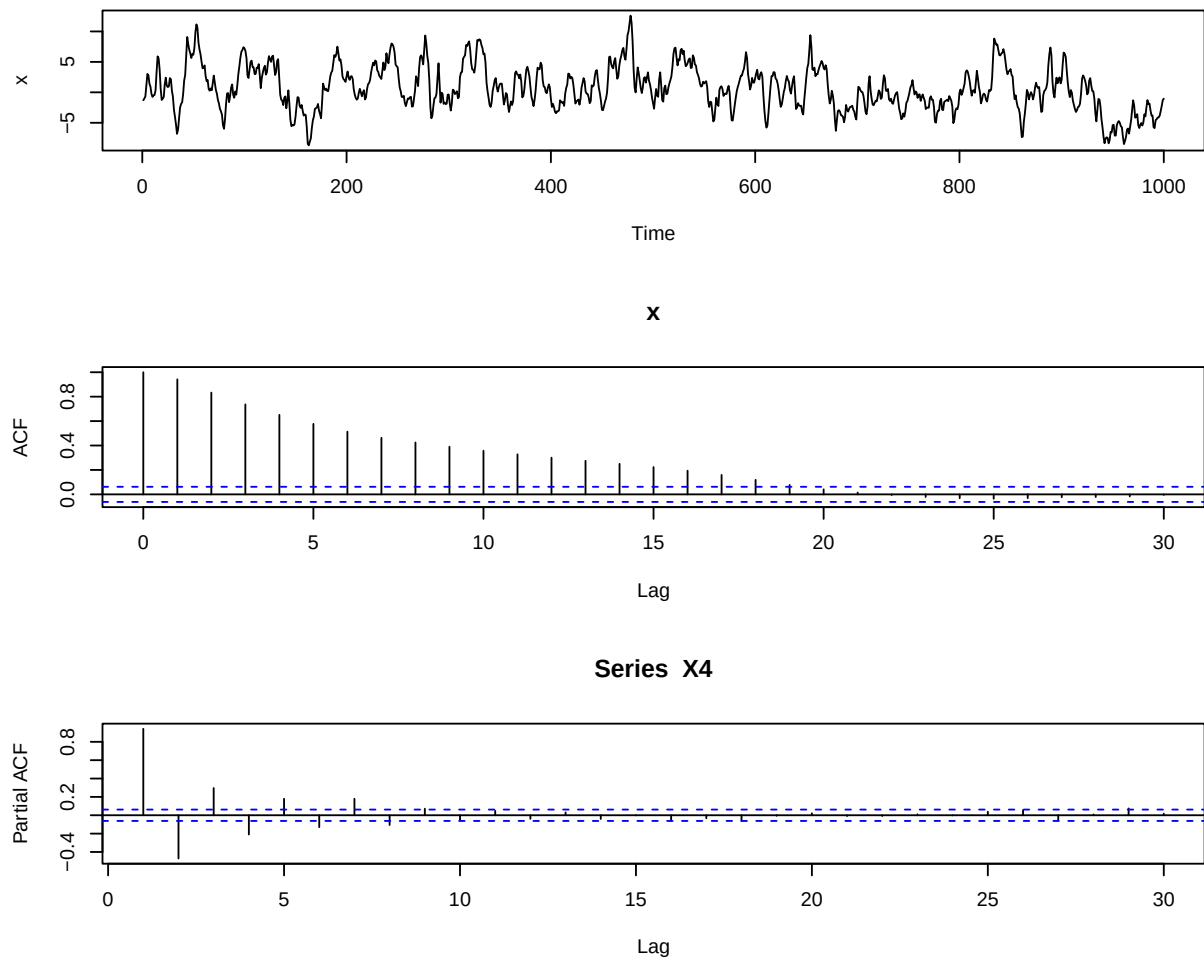
2026-03-14

## Data Loading

```
serie4= read.table("serie4.txt", header=T); X4= ts(serie4)
```

## Exercise 1 - Visual analysis and order estimation

```
par(mfrow=c(3,1)); plot(X4); acf(X4); pacf(X4)
```



From the ACF and PACF plots, neither function exhibits a clear cutoff; both gradually decay. This behavior is characteristic of an ARMA( $p,q$ ) process rather than a pure AR( $p$ ) or MA( $q$ ) process. Therefore, we consider ARMA models for several combinations of  $p$  and  $q$ .

```
pmax <- 5
qmax <- 5

AIC_mat <- matrix(NA, pmax, qmax)
BIC_mat <- matrix(NA, pmax, qmax)

for (p in 1:pmax) {
  for (q in 1:qmax) {

    fit <- arima(X4, order=c(p,0,q))

    AIC_mat[p,q] <- AIC(fit)
    BIC_mat[p,q] <- BIC(fit)
  }
}
```

```

}
}

knitr::kable(AIC_mat, caption="AIC values for ARMA(p,q)")

```

Table 1: AIC values for ARMA(p,q)

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| 2718.499 | 2720.042 | 2721.896 | 2723.774 | 2725.039 |
| 2720.029 | 2721.991 | 2723.791 | 2724.946 | 2726.391 |
| 2721.920 | 2723.751 | 2723.008 | 2724.948 | 2728.404 |
| 2723.756 | 2725.602 | 2726.592 | 2728.584 | 2732.890 |
| 2724.886 | 2726.486 | 2728.158 | 2722.105 | 2730.819 |

```

knitr::kable(BIC_mat, caption="BIC values for ARMA(p,q)")

```

Table 2: BIC values for ARMA(p,q)

|          |          |          |          |          |
|----------|----------|----------|----------|----------|
| 2738.130 | 2744.581 | 2751.343 | 2758.129 | 2764.301 |
| 2744.568 | 2751.438 | 2758.146 | 2764.208 | 2770.561 |
| 2751.366 | 2758.106 | 2762.270 | 2769.118 | 2777.482 |
| 2758.111 | 2764.865 | 2770.762 | 2777.661 | 2786.875 |
| 2764.148 | 2770.656 | 2777.236 | 2776.090 | 2789.712 |

```

which(AIC_mat == min(AIC_mat), arr.ind=TRUE)
which(BIC_mat == min(BIC_mat), arr.ind=TRUE)
which(AIC_mat==sort(AIC_mat)[2], arr.ind=TRUE)
which(BIC_mat==sort(BIC_mat)[2], arr.ind=TRUE)

```

```

      row col
[1,]    1   1
      row col
[1,]    1   1
      row col
[1,]    2   1
      row col
[1,]    2   1

```

The minimum AIC and BIC values correspond to the models ARMA(1,1) and ARMA(2,1).

## Exercice 2 - Maximum Likelihood parameter estimation

```
fit11 <- arima(X4, order=c(1,0,1), method="ML")
fit21 <- arima(X4, order=c(2,0,1), method="ML")

fit11
```

Call:  
 arima(x = X4, order = c(1, 0, 1), method = "ML")

Coefficients:

|      | ar1    | ma1    | intercept |
|------|--------|--------|-----------|
|      | 0.8816 | 0.8780 | 0.6334    |
| s.e. | 0.0149 | 0.0154 | 0.4658    |

sigma^2 estimated as 0.8768: log likelihood = -1355.25, aic = 2718.5

```
fit21
```

Call:  
 arima(x = X4, order = c(2, 0, 1), method = "ML")

Coefficients:

|      | ar1    | ar2    | ma1    | intercept |
|------|--------|--------|--------|-----------|
|      | 0.8592 | 0.0246 | 0.8835 | 0.6277    |
| s.e. | 0.0359 | 0.0358 | 0.0167 | 0.4757    |

sigma^2 estimated as 0.8764: log likelihood = -1355.01, aic = 2720.03

## Exercise 3 - Model Equation

The equations of the two selected models are:

$$ARMA(1,1) : X_t = 0.6334 + 0.8816X_{t-1} + 0.8780\epsilon_{t-1} + \epsilon_t$$

$$ARMA(2,1) : X_t = 0.6277 + 0.8592X_{t-1} + 0.0246X_{t-2} + 0.8835\epsilon_{t-1} + \epsilon_t$$

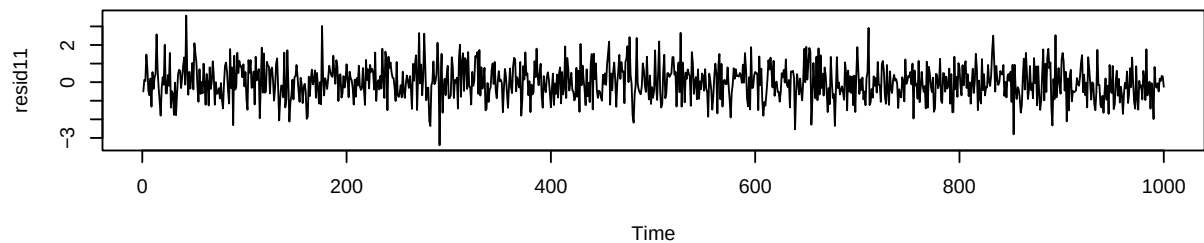
## Exercise 4 - Residual diagnostics

To determine whether the models are correctly specified, we analyze the residuals. If the model adequately captures the dependence structure of the series, the residuals should behave like **white noise**.

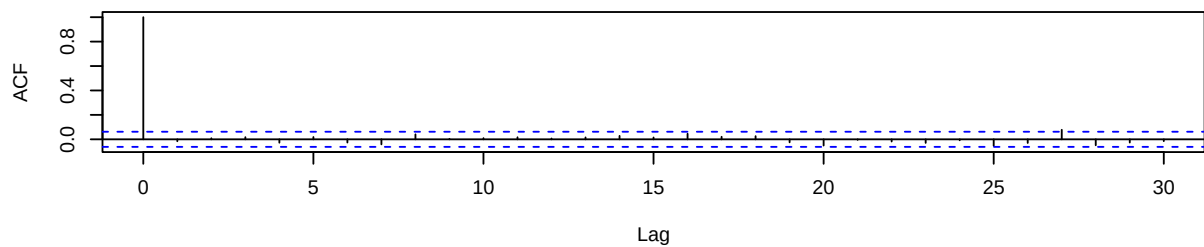
We therefore examine: - the residual time series - the ACF and PACF of the residuals - Ljung-Box tests for remaining autocorrelation.

## Model 1 diagnostics

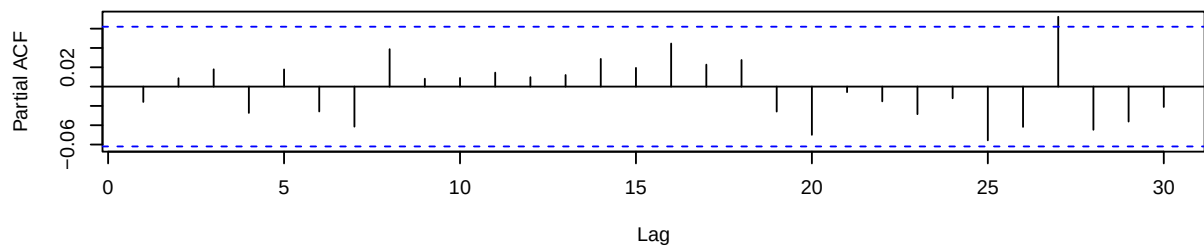
```
par(mfrow=c(3,1))
resid11=residuals(fit11)
plot(resid11)
acf(resid11)
pacf(resid11)
```



Series resid11



Series resid11



The residuals fluctuate randomly around zero and no clear pattern is visible. The ACF and PACF show no significant spikes outside the confidence bands, indicating the absence of remaining auto-correlation.

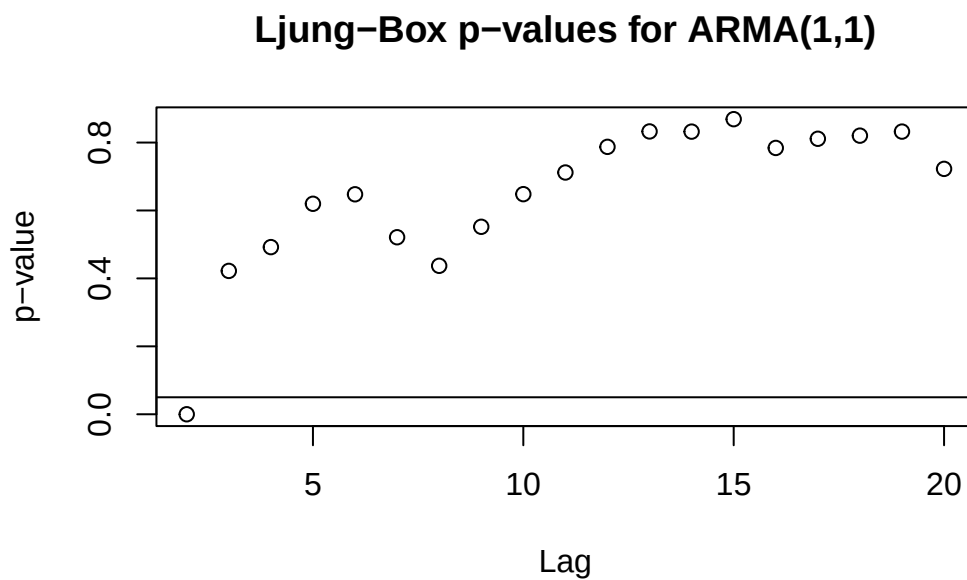
## Ljung- Box Test

```
pval1 <- numeric(19)

for(i in 2:20){
  pval1[i-1] <- Box.test(resid11, lag=i,
                        type="Ljung",
                        fitdf=2)$p.value
}

plot(2:20, pval1,
     main="Ljung-Box p-values for ARMA(1,1)",
     xlab="Lag", ylab="p-value")

abline(h=0.05)
```

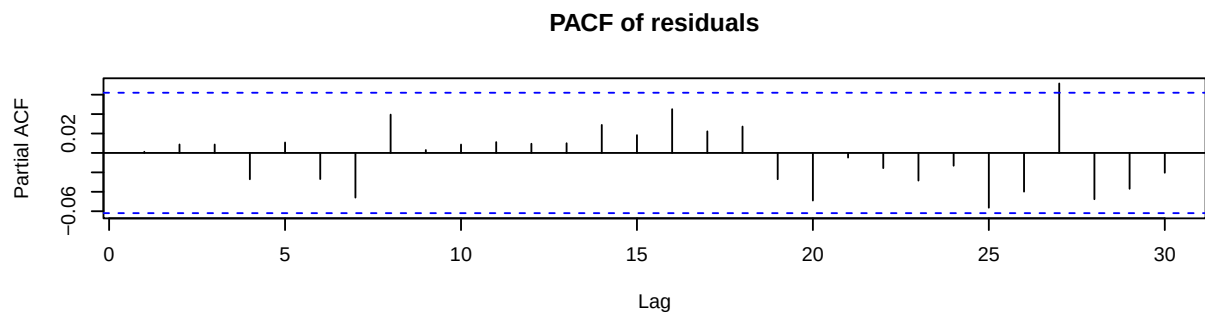
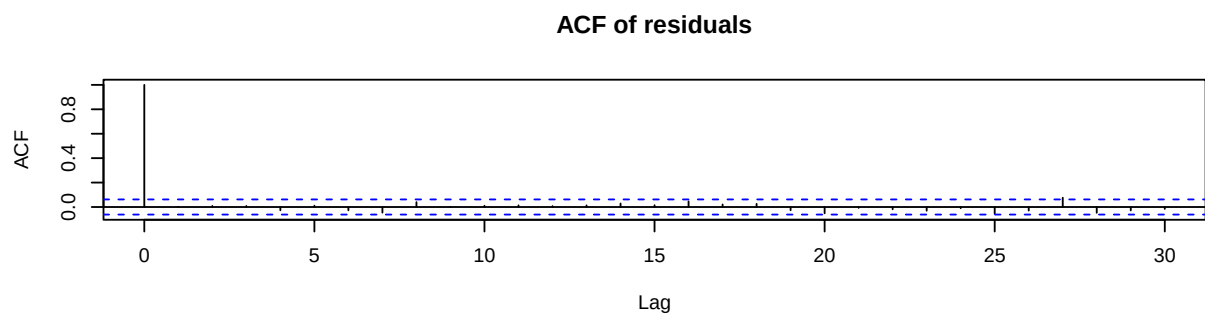
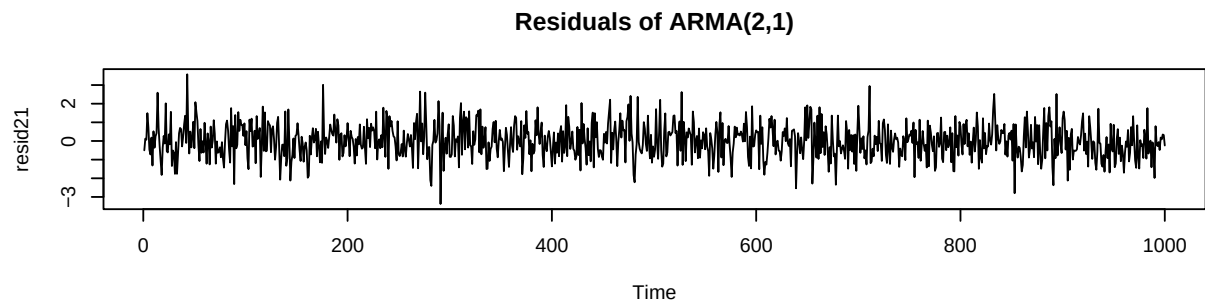


Most p-values are above the significance level (0.05), suggesting that the residuals do not exhibit significant autocorrelation. Therefore the ARMA(1,1) model appears to adequately capture the correlation structure of the data.

## Model 2

```
par(mfrow=c(3,1))
resid21 <- residuals(fit21)
plot(resid21, main="Residuals of ARMA(2,1)")
```

```
acf(resid21, main="ACF of residuals")
pacf(resid21, main="PACF of residuals")
```



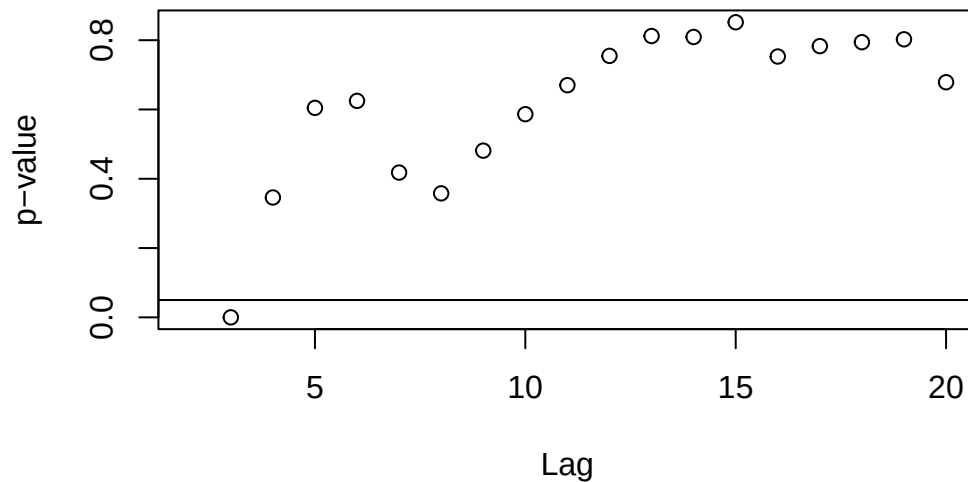
```
pval2 <- numeric(19)

for(i in 2:20){
  pval2[i-1] <- Box.test(resid21, lag=i,
                        type="Ljung",
                        fitdf=3)$p.value
}

plot(2:20, pval2,
     main="Ljung-Box p-values for ARMA(2,1)",
     xlab="Lag", ylab="p-value")

abline(h=0.05)
```

### Ljung-Box p-values for ARMA(2,1)



Similarly, the residual diagnostics indicate that the ARMA(2,1) model does not leave significant autocorrelation in the residuals. Thus this model also appears to be correctly specified.

Between the two models, the **ARMA(1,1)** model is more parsimonious because it contains fewer parameters than the *ARMA(2,1)* model.

Since both models adequately capture the correlation structure of the data, the ARMA(1,1) model is preferred.

## Exercise 5 - One-step-ahead prediction

To compare the predictive performance of the two models, we compute one-step-ahead forecasts using a rolling window approach.

```
OneAhead <- function(x, order, K = 0.9){  
  
  n <- length(x)  
  ntrain <- floor(K*n)  
  
  preds <- numeric(n)  
  
  for(i in ntrain:(n-1)){  
    fit <- arima(x[1:i], order=order)  
    preds[i+1] <- predict(fit, n.ahead=1)$pred  
  }  
  
  idx <- (ntrain+1):n  
  pred <- preds[idx]
```

```

errors <- x[idx] - pred

list(pred=pred,
     errors=errors,
     mspe=mean(errors^2))
}

```

```

OAP11 <- OneAhead(X4, order=c(1,0,1))
OAP21 <- OneAhead(X4, order=c(2,0,1))

OAP11$mspe
OAP21$mspe

```

```

[1] 0.7929316
[1] 0.7904022

```

This confirms our hypothesis made earlier. These two models are almost identical in terms of predicting the data.

For further analysis, we plot the *actual* vs the *predicted*.

```

n <- length(X4)
ntrain <- floor(0.9*n)

actual <- X4[(ntrain+1):n]

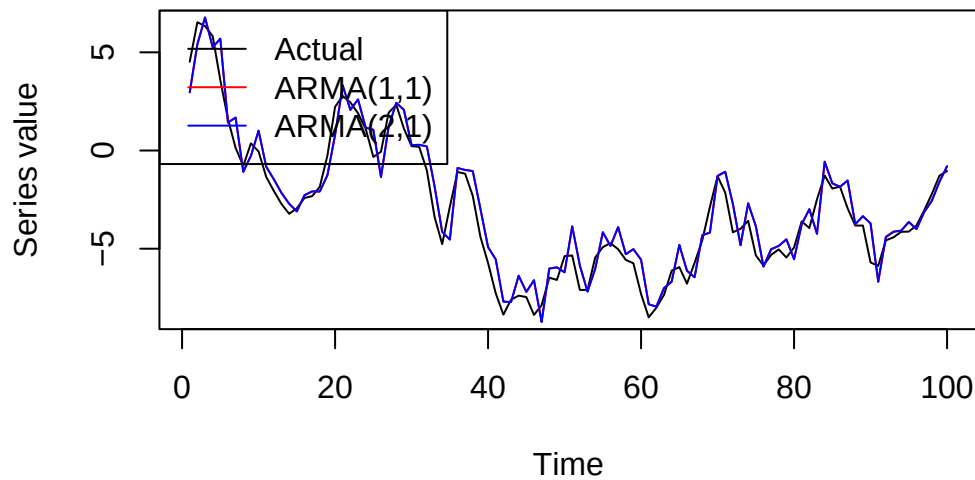
plot(actual,
     type="l",
     col="black",
     main="One-step-ahead predictions",
     ylab="Series value",
     xlab="Time")

lines(OAP11$pred, col="red")
lines(OAP21$pred, col="blue")

legend("topleft",
     legend=c("Actual", "ARMA(1,1)", "ARMA(2,1)"),
     col=c("black", "red", "blue"),
     lty=1)

```

## One-step-ahead predictions



Again, we observe the same thing, the one step ahead predictions of the ARMA(1,1) and the ARMA(2,1) are intertwined. They make the same predictions

## Exercise 6

Both models appear to adequately capture the correlation structure of the data. However, the ARMA(1,1) model is more parsimonious. Therefore, the **ARMA(1,1)** model is selected as the most appropriate model for the series.

## Exercise 7 - Forecasting

```
forecast <- predict(fit11, n.ahead = 10)

# last 50 observations
last50 <- tail(X4, 50)

plot(last50,
      xlim = c(time(last50)[1], time(last50)[50] + 10),
      ylim = range(c(last50, forecast$pred)),
      main = "10-step forecast with ARMA(1,1)", type="l",
      ylab = "Series",
      col = "black")

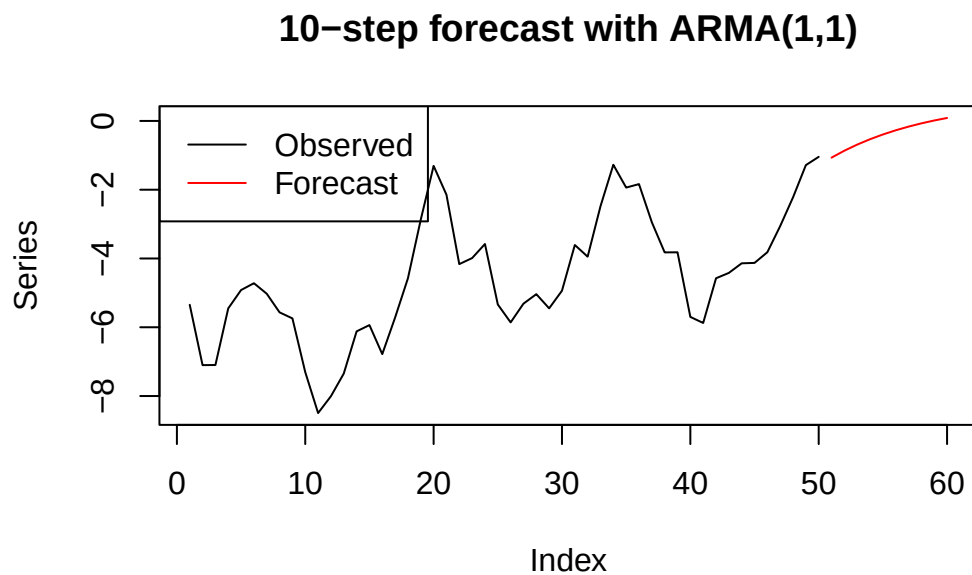
lines(seq(time(last50)[50] + deltat(X4),
```

```

    by = deltat(X4),
    length.out = 10),
forecast$pred,
col = "red")

legend("topleft",
      legend = c("Observed", "Forecast"),
      col = c("black", "red"),
      lty = 1)

```



The figure shows the last 50 observations of the series together with the 10-step-ahead forecasts obtained from the selected ARMA(1,1) model.